MSG No. 70.531 $Fd'd'd'$ [Type III, orthorhombic]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}$	1	1
2	$\{2_{100}\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix}1&0&0&0\\0&-1&0&\frac{3}{4}\\0&0&-1&\frac{3}{4}\end{bmatrix}$	1	1
3	$\{2_{010}\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix}-1&0&0&\frac{3}{4}\\0&1&0&0\\0&0&-1&\frac{3}{4}\end{bmatrix}$	1	1
4	$\{2_{001}\|\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix}-1&0&0&\frac{3}{4}\\0&-1&0&\frac{3}{4}\\0&0&1&0\end{bmatrix}$	1	1
5	$\{-1'\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&0\\0&0&-1&0\end{bmatrix}$	−1	−1
6	$\{m_{100}'\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix}-1&0&0&0\\0&1&0&\frac{1}{4}\\0&0&1&\frac{1}{4}\end{bmatrix}$	−1	−1
7	$\{m_{010}'\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix}1&0&0&\frac{1}{4}\\0&-1&0&0\\0&0&1&\frac{1}{4}\end{bmatrix}$	−1	−1
8	$\{m_{001}'\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix}1&0&0&\frac{1}{4}\\0&1&0&\frac{1}{4}\\0&0&-1&0\end{bmatrix}$	−1	−1
9	$\{1\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
10	$\{2_{100}\|0\frac{1}{4}\frac{1}{4}\}$	$\begin{bmatrix}1&0&0&0\\0&-1&0&\frac{1}{4}\\0&0&-1&\frac{1}{4}\end{bmatrix}$	1	1
11	$\{2_{010}\|\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix}-1&0&0&\frac{3}{4}\\0&1&0&\frac{1}{2}\\0&0&-1&\frac{1}{4}\end{bmatrix}$	1	1
12	$\{2_{001}\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&\frac{3}{4}\\0&-1&0&\frac{1}{4}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
13	$\{-1'\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	−1	−1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{\mathrm{m}_{100}{}'\|0\frac{3}{4}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	-1	-1
15	$\{\mathrm{m}_{010}{}'\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	-1	-1
16	$\{\mathrm{m}_{001}{}'\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	-1
17	$\{1\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
18	$\{2_{100}\|\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
19	$\{2_{010}\|\frac{1}{4}0\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{4} \end{bmatrix}$	1	1
20	$\{2_{001}\|\frac{1}{4}\frac{3}{4}\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	1	1
21	$\{-1'\|\frac{1}{2}0\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	-1
22	$\{\mathrm{m}_{100}{}'\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	-1	-1
23	$\{\mathrm{m}_{010}{}'\|\frac{3}{4}0\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}$	-1	-1
24	$\{\mathrm{m}_{001}{}'\|\frac{3}{4}\frac{1}{4}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	-1
25	$\{1\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
26	$\{2_{100}\|\frac{1}{2}\frac{1}{4}\frac{3}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	1	1
27	$\{2_{010}\|\frac{1}{4}\frac{1}{2}\frac{3}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{3}{4} \end{bmatrix}$	1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
28	$\{2_{001}\|\frac{1}{4}\frac{1}{4}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{4} \\ 0 & -1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
29	$\{-1'\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	−1
30	$\{\mathrm{m}_{100}'\|\frac{1}{2}\frac{3}{4}\frac{1}{4}\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	−1	−1
31	$\{\mathrm{m}_{010}'\|\frac{3}{4}\frac{1}{2}\frac{1}{4}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} \end{bmatrix}$	−1	−1
32	$\{\mathrm{m}_{001}'\|\frac{3}{4}\frac{3}{4}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} \\ 0 & 1 & 0 & \frac{3}{4} \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	−1